industries, transforming them from specialized military devices into mainstream tech-nological assets that support economic growth, sustainability, and digital transformation.

Figure 1.6 illustrates a quadcopter-based fourth-generation drone equipped with multi-sensor modules, GPS, LiDAR, and on-board cameras. The figure highlights the integration of a flight controller and an AI-assisted navigation system, enabling semi-autonomous operation. Various application domains, such as agriculture, construction, environmental monitoring, and logistics, are represented through icons surrounding the drone, showcasing its multifunctional commercial role and technological versatility.

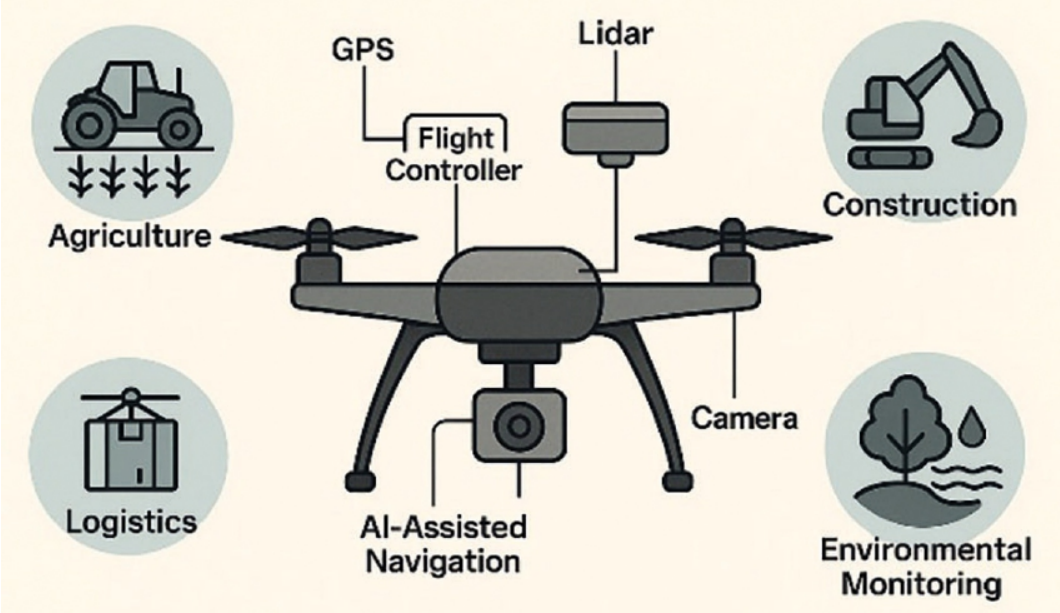


Figure 1.6 Fourth-generation drone semi-autonomy and commercial applications. ⏎

1.4.5 Fifth generation: Full autonomy, AI, and swarm technology

The fifth generation of drone technology represents the pinnacle of unmanned aerial system evolution, characterized by full autonomy, artificial